

## Chapter 142

# Grade Management

The Color page offers numerous ways of saving grades, copying grades, creating and altering groups of clips to share grades, rippling grades and adjustments, and managing multiple versions of grades.

All of these procedures enable you to work faster by leveraging work you've already done to apply to other clips that can benefit from the same adjustments, or by applying changes across multiple clips all at once.

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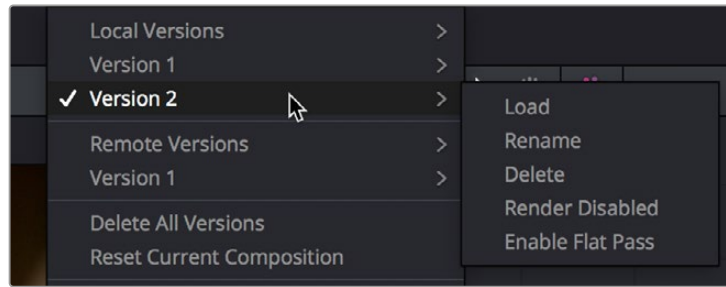
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## Using Versions to Manage Grades

Working quickly requires a mastery of the numerous grade management features built into DaVinci Resolve. These features are designed to help you copy and ripple grades to individual clips, among manually defined groups of clips via the Node Editor's Clip mode, or among automatically linked clips that share the same source clip in the Media Pool (when using Remote versions). Multiple versions of grades are supported for every clip, as are various options for previewing, overwriting, and appending these versions.

### Choosing to Copy Grades Manually or Automatically Using Versions

Each grade you apply to a clip is a version. Each clip can have multiple versions, although only one version can be applied at a time. By default, the first grade applied to every clip in a timeline is a local version named "Version 1." A clip's version name and/or number appears underneath the clip thumbnail, to the right. You can also see the version name, along with a list of all the other versions that are currently available to that clip, by right-clicking a clip in the Timeline and looking for the version names that appear underneath the Local Versions submenu.



Selecting versions of grades from the Timeline thumbnails

**NOTE:** Double-clicking the version name underneath a clip's thumbnail in the Timeline cycles the version names, the clip names, and the codec used by each clip.

There are two different types of versions in DaVinci Resolve, each of which provides a different method of grade management and linking among the clips and timelines of a project.

**Local versions:** The default mode of grade management. Clips using local versions are unlinked, so each clip has its own grade which is not shared with any other clips in any other timelines. Changing the grade of a clip using a local version has no effect on any other clip in that project.

**Remote versions:** An alternate mode of grade management that you can enable. All clips using remote versions that share the same source clip in the Media Pool are automatically linked, in all timelines, and share the same set of grades.

If you use local versions to grade all your clips, you don't have to worry about one clip's grade being accidentally copied to other clips that happen to come from the same file in the Media Pool, which can make it easier to work. On the other hand, you need to manually copy every grade that you want to duplicate, even among clips that clearly come from the same take or angle. You can always create a group to share a grade among clips using local versions, but that's an extra step.

Switching to use remote versions, instead, puts you into a mode where clips using remote versions that share the same media file are automatically linked together. This gives you a convenient shortcut for grading your program, since a grade that's applied to one linked clip is automatically copied to every other clip it's linked to, which can be convenient when grading a series of headshots appearing throughout a program that come from the same interview take. Another use of remote versions is when you're importing a new edit of an already graded timeline. Using remote versions, you can set your project up so that new incoming timelines are automatically relinked to the previous timeline's grades.

The only disadvantage to using remote versions is that when you find you need to make individual adjustments to clips that are linked, it's an extra step to either create another version, or switch that clip to use a local version, in order to keep its adjustment separate. This will be discussed in more detail later in this chapter.

## Mixing Versions

Each clip is capable of having multiple local and remote versions at the same time; you get to choose which to apply. Which type of versioning is best depends on the type of project you're working on, the way in which the media was shot, and how you like to work. The following sections explain the differences in workflow.

## Using Local Versions by Default

Since local versions are the default method of grading when you first create a new project (unless you've edited your settings presets), you don't need to do anything to enable this mode at first. As the name implies, local versions are local to the Timeline in which they appear, so local versions are not rippled to linked clips, and they're not shared among different timelines. This makes it easy to avoid accidentally having grades copied when you don't want them to be, but you lose the conveniences that automatic linking can provide.

However, you can always switch the clips in a timeline to use remote versions if you change your mind, either individually, or all together.

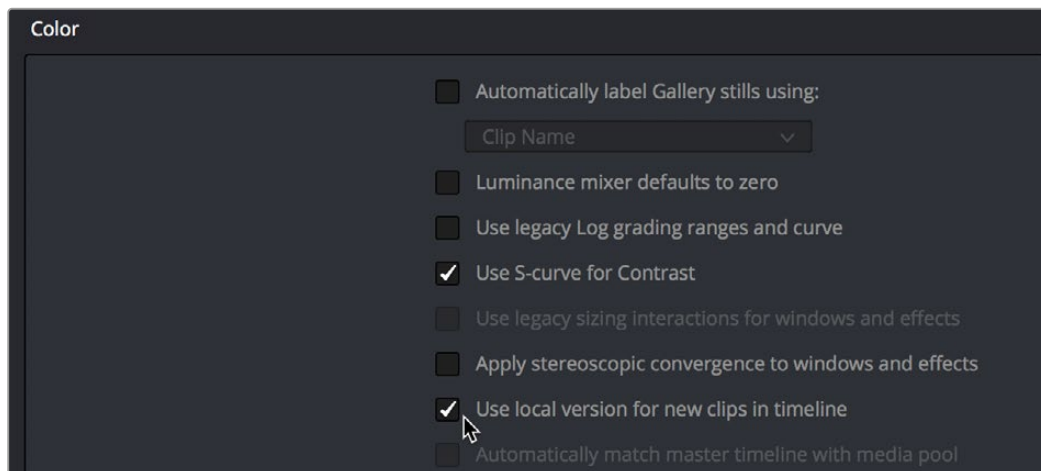
## Using Remote Versions to Enable Automatic Linking

Remote versions let all clips from the same source media file in the Media Pool share their grades automatically, either within a single timeline, or across multiple timelines in the same project. However, to enable the convenience of remote grades, you need to put DaVinci Resolve into this mode of working using the Settings window.

### To enable remote grades to be used:

- Open the Project Settings, open the General Options panel, and turn off "Use local version for new clips in timeline" in the Color section.

Turning this option off only affects clips that are added to Timelines from that point onward. You can keep track of which clips use remote versions by the (R) that appears when you double-click the area underneath the thumbnail to hide the codec name in the Thumbnail timeline.

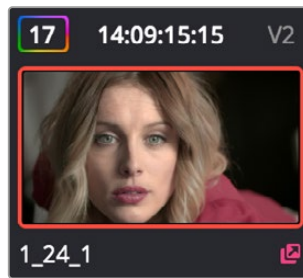


Turning off "Use local version for new clips in timeline" to enable remote grades to be used by default

## How Automatic Linking Works

When clips using remote versions have been added to a timeline, any other clips in any timeline that

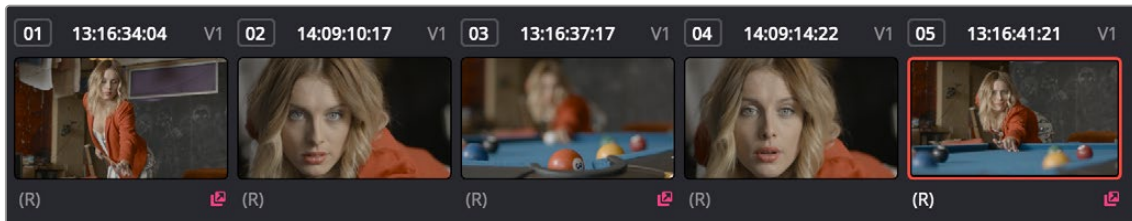
- a) use remote versions also, and
- b) refer to the same file in the Media Pool are linked. Selecting a clip using a remote grade in the Color page that is automatically linked to one or more clips displays a small "linked" badge at the right of that clip's timecode in the Thumbnail timeline.



The linking arrow shows this clip has the same source as the one selected on the Timeline

**TIP:** The Timeline Filtering drop-down in the Color Page toolbar has an option named “Common Media Pool Clips” that only shows clips in the Timeline that are linked to the currently selected clip.

Adjustments that you apply to a remote version of one linked clip are automatically rippled to every other clip it’s linked to, which can save you an enormous amount of time when you’re getting started with a new program. For example, if every reverse angle from the same take in a particular angle of coverage is automatically linked using remote versions, then the grade you apply to one reverse angle clip will ripple to every other reverse angle clip throughout the Timeline.



Timeline with three automatically linked clips shown, and codec names hidden to reveal the (R) that shows these clips are using remote grades

So long as remote versions are enabled, there are other ways that linked clip relationships are formed. If, after conforming an AAF, XML, or EDL, you used the Split Clips button in the Color page to split one conformed clip into many, each of the clips you split would be linked, since they too would share the same source media in the Media Pool.

Finally, automated linking also occurs for clips that appear in multiple timelines that also use remote versions. As a result, the grade you apply to one linked clip is automatically rippled to every other clip it’s linked to.

**NOTE:** Media management or media consolidation operations that split large source media files into many individual media files will defeat automatic linking within the same timeline, since each clip will be conformed to its own individual media file.

## Starting with Remote Versions, Then Switching to Local Versions

It's possible to combine the best aspects of remote and local versions into a single workflow, taking advantage of your ability to freely switch from one to the other. Since remote versions make it easy to copy grades among clips that are similar, you can switch to using remote versions first, and then grade your way through the Timeline until you get to the point where you need to start making much more specific changes to individual clips. Then, it's easy to either switch each linked clip that needs individual adjustment to use a local version, or to use the Copy Remote Grades To Local command (described later) to copy the current remote version of each and every clip to a local version, at which point you can make all the specific tweaks you need to without worrying about changes being accidentally copied.

## Creating a Master Timeline

DaVinci Resolve version 9 and earlier would automatically create a Master Timeline whenever you added clips to the Media Pool. This changed in versions 10 and later, which by default have no Master Timeline. However, if you want to create a Master Timeline in order to use it as you would have before, this is easy to do.

If you want a Master Timeline to have a single timeline that always contains all clips currently in the Media Pool, there's a way you can create one. However, you need to do it immediately upon creating a new project, before adding any media to the Media Pool. Once you've added one or more clips to the Media Pool, the option you need to do so will be disabled.

### To create a new Master Timeline:

- 1 Create a new project, open the General Options panel of the Project Settings, and turn on the "Automatically match master timeline with media pool" checkbox in the Color section. If you also want all clips to use Remote versions as you grade by default as in previous versions of DaVinci Resolve, you can turn off the "Use local version for new clips in timeline" checkbox.
- 2 Click Save.
- 3 Open the Edit page, and choose File > New Timeline (Command-N).
- 4 When the New Timeline Properties window appears, turn the Empty Timeline checkbox off, and click Create New Timeline.

Now, in addition to the new Timeline, a Master Timeline appears in the Timeline list.

Once created, the Master Timeline contains every clip in the Media Pool of the current project. If you color correct the clips in the Master Timeline, you'll find there's only one set of versions available, found underneath the Local submenu of the Color Page timeline's contextual menu. It's important to understand that local versions within the Master Timeline are in fact the remote versions found in every other timeline of that project.

In fact, it would be fair to say that the remote versions found in other timelines are actually the Master Timeline's versions. When you grade a clip in the Master Timeline, those grades ripple out to every other instance of that clip, in every other timeline within that project, via remote versions.

Sharing of remote versions among the Master Timeline and newly conformed timelines is why you can grade a collection of clips that you import into DaVinci Resolve without any editorial structure, grade and output offline media, and then reimport a project file that relinks to the original clips along with their grades. It's also why you can grade clips in one timeline, and then import additional re-edited timelines via AAF, XML, or EDL, which automatically inherit all remote versions of grades that were created in previous timelines.

## Differentiating Clips with Individual Versions

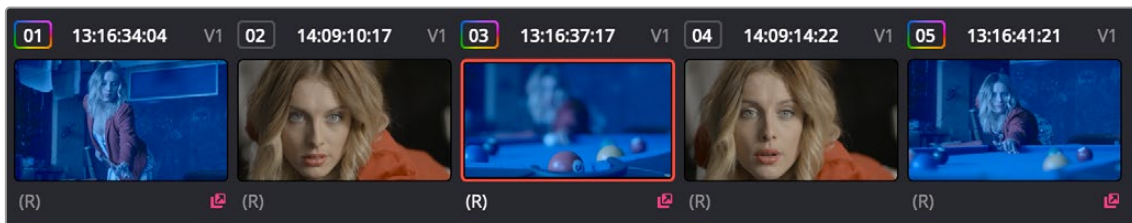
If you've enabled DaVinci Resolve to add clips with remote versions, there are times when you may want to suspend automatic linking. As convenient as automatic grade rippling among linked remote versions is, there are many instances where you may need to stop it. For example, if the Media Pool contains a source media file with the entire content of an interview, then every clip that has been conformed to that source media will be linked, which is ordinarily good since you'd expect they'd all share the same grade. However, if the DP fiddled with the exposure of the camera in the middle of the interview, such that some clips are lighter and others are darker, you may find yourself needing to make different adjustments to clips from different parts of the interview.

One way of doing this, which can be useful in timelines where you only need to make a few of these kinds of changes, is to create a new remote version for every clip to which you need to make an individual adjustment. Since each version is its own grade, and differently named versions don't link to one another, this is a simple solution.

### To suspend linking by creating a new remote version:

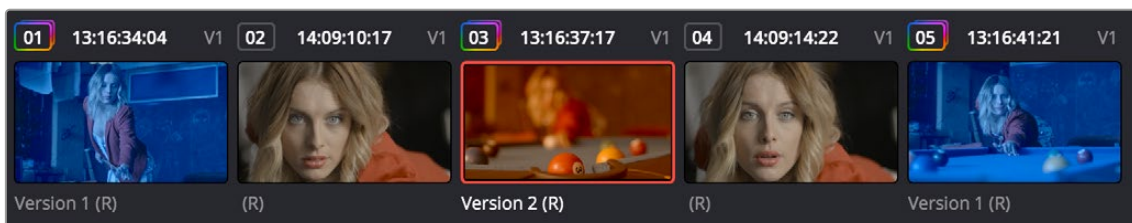
- 1 Move the playhead to the clip you need to alter individually.
- 2 Do one of the following:
  - Choose Color > Grade Version > Add (Command-Y).
  - Right-click the thumbnail in the Timeline, and choose Remote Versions > Create New Version.
- 3 Grade the new version that's appeared.

While this method works well, keep in mind that multiple linked clips using the same version number will always be linked with one another. In other words, suppose Clip 1, Clip 3, and Clip 5 are linked using the default Version 1, which has a strong blue grade applied.



Three clips automatically linked, sharing a blue grade

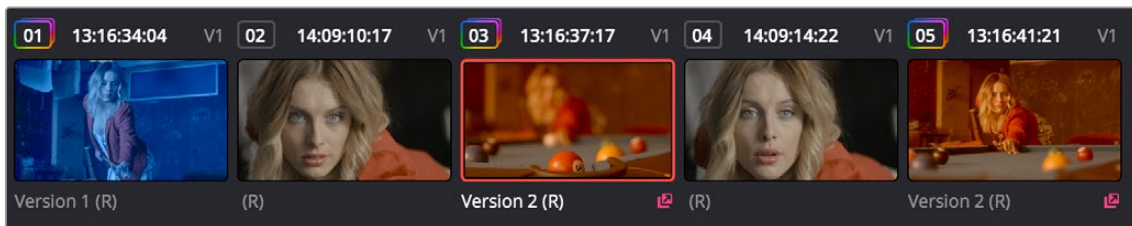
Then, you set Clip 3 to use Version 2 with a different grade, tinted red. Clip 3 is now unlinked from Clips 1 and 5.



Setting the second of the three clips to use another version with a different remote grade sets it apart

However, switching Clip 5 to also use the red-tinted Version 2 means that Clip 3 and Clip 5 are now linked together, but Clip 1 is no longer linked.





Setting the third of these clips to share the same remote version as the second clip now links those two together, omitting the first clip

Every new remote version you create, and every change you make, becomes available to all other clips in all other timelines that comes from the same source Media Pool clip as the shot you're working on. However, any clip can use any version, and which version is used by a linked clip is not rippled.

For example, Timeline 1 contains a clip named Max CU, which has three remote versions. Timeline 5 also contains Max CU, which is currently set to use Version 2. If you open Timeline 1 and add one more remote version to Max CU, that new version will also be available to Max CU in Timeline 5, but it will still be set to use Version 2.

## Switching Clips Between Local and Remote Versions

You can also suspend remote version grade linking by switching individual clips to use local versions. This makes it easy to create a situation where some clips are linked, and other clips are not. Keep in mind that each clip has both local and remote versions, so you can switch from one to the other without losing anything.

### To suspend linking by setting individual clips to use a local version:

Right-click a clip thumbnail, choose the submenu of the corresponding remote version you want to copy, and choose Copy to Local. The remote version is copied to a local version, which now appears as the current version that's used.

You can also set every clip in the entire timeline to use local versions. If you switch an entire timeline to local versions, then no clip in that timeline will be linked to any other unless you create a group (covered later in this chapter). This is the default state, but it's useful even when you've switched to using remote versions when you want to grade one timeline differently from the others in a project, for example when doing a trim pass for another video or stereo 3D format, or when grading another cut, such as a trailer, that uses the same media but requires a different look.

### To suspend linking by switching from remote versions to local versions:

Right-click any clip thumbnail, and choose one of the following commands;

**Use Local Grades:** Switches all clips to their local versions. If there are already local versions defined for each clip, those will appear. If there are no local versions yet defined, each clip will be ungraded.

**Copy Remote Grades to Local:** Copies the currently assigned remote version of each clip to a local version.

Because Undo is used only for individual clip operations, the Use Local Grades and Copy Remote Grades To Local commands cannot be undone. However, switching a timeline from remote to local versions is not a one-way trip. You can select Use Remote Grades to switch every clip in a timeline back to the remote versions at any time.

### To switch a timeline back to using remote versions:

Right-click any clip thumbnail, and choose Use Remote Grades.



Furthermore, you can also copy the local grades you made to be remote grades, for instances where you started grading with local grades, and you want to switch to remote grading using your local grades as a starting point. Be aware that when you do this in a timeline with many clips that share the same source media, the grade of the very last shared clip in the Timeline will be the only remote grade that's used for those auto-linked clips.

**To copy each clip's local grade to a remote grade for every clip in the current timeline:**

Right-click any clip thumbnail, and choose Copy Local Grades to Remote.

**If you're switching back and forth between local and remote versions, keep the following rules in mind:**

- If you select Use Remote Grades, change your remote grades, and then Copy Remote Grades To Local on that timeline again, you'll overwrite all of your previously graded local versions. This is a good workflow if you don't like what you've done with your local versions, and you want to start over by recopying the remote versions.
- If you select Use Remote Grades, change your remote grades, and then select Use Local Grades, you'll return to your previously graded local versions, as they were before you selected Use Remote Grades. This is a good workflow if you just want to switch back to your remote versions in order to copy the remote versions of specific clips to local versions.

## Working with Versions

Each clip can have as many local and remote versions as you like. There are commands for creating, renaming, and deleting the versions that are available to each clip when using the pointer or keyboard shortcuts.

Additionally, DaVinci Resolve 15 introduced the ability to batch create, rename, load, and delete versions for multiple clips that are selected at the same time.

**To create a new local or remote version quickly for one or more selected clips:**

- Choose Color > Grade Version > Add (Command-Y).

A new version is created, identical to the type of version that was previously selected (remote or local), named "Version x" where x is the number of the versions that have been created. All keyframes and motion tracking are copied from the previously selected version to the new one.

**To create a new version for one or more selected clips, with the option to set a name and whether to copy keyframes:**

- 1 Right-click one of the selected clip thumbnails, and choose one of the following:
  - Local Versions > Create New Version
  - Remote Versions > Create New Version
- 2 Type a name for the version in the dialog that appears, and choose either "Copy all marks" or "Copy the first mark only," depending on whether or not you want to copy all the keyframes into the new version.

**NOTE:** Motion tracking will be copied regardless of whether "Copy all marks" is selected or not.

- 3 Optionally, you can choose one of the preset version names from the drop-down list. These names are defined in the Versions section of the General Options panel of the Project Settings.
- 4 Click OK.

**To delete a specific version for one or more selected clips:**

- Right-click one of the selected clip thumbnails, choose the submenu of the remote version you want to delete (it can't be the currently selected version), and choose Delete.

**To delete all versions for one or more selected clips:**

- Right-click one of the selected clip thumbnails, and choose one of the following commands from the contextual menu corresponding to the scope of the versions you want to delete;
  - Local Versions > Delete All Versions
  - Remote Versions > Delete All Versions
  - Delete All Versions

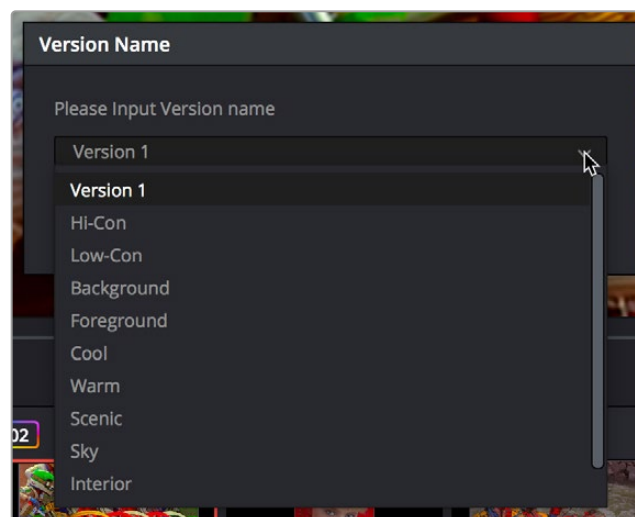
All versions are deleted as specified, except for the currently selected version, which is now the only one remaining.

**To load a particular remote or local version for one or more selected clips:**

- Right-click one of the selected clip thumbnails, choose the submenu of the corresponding remote version you want to copy, and choose Load.

**To rename the version of one or more selected clips:**

- 1 Right-click one of the selected clip thumbnails, choose the submenu of the corresponding remote version you want to rename, and choose Rename.
- 2 In the Version Name dialog, type a name into the text field. Optionally, you can choose one of the preset version names from the drop-down list. These names are defined in the Versions section of the General Options panel of the Project Settings.

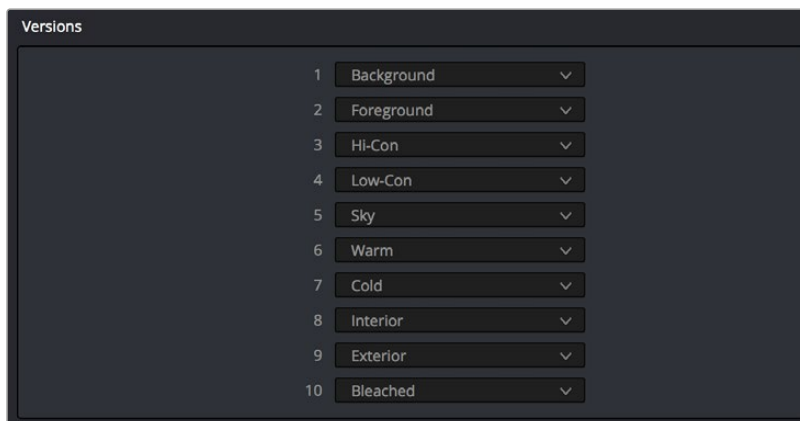


Selecting a Version name from the dropdown list, the names on the list come from assignments made using the Versions section of the General Options panel of the Project Settings

- 3 Click OK.

That name now appears underneath the clip thumbnail in the Timeline.

If you're going to use predefined names for versions, you need to define these in the Versions group of the General Options panel of the Project Settings. Ten drop-down menus let you choose from preset version names, or enter your own. These names will appear in the drop-down menu of the Version Name dialog.



The Versions section of the Project Settings

## The Importance of “Version 1”

There are specific operations in DaVinci Resolve where “Version 1,” otherwise known as “the default version,” is the only version that’s used, regardless of how many other versions are available, or which version was previously selected. The default version can be named anything you want, but whatever its name, the default version is the first version in the list, and it’s important. Here are some examples:

**Importing additional timelines that use the same clips:** In this case, if you’re importing timelines and “Use local version for new clips in timeline” is disabled in the Color section of the General Options panel of the Project Settings, then only the default version of each clip is relinked.

**Whenever you switch a timeline between local and remote versions:** In this case, every clip in the Timeline will be switched to the default version, regardless of what you had selected previously.

Consequently, if you plan to reconform your clips to a newer version of the edit that you’re importing, or pursue other workflows that are off the beaten path, it’s a good idea to make sure, no matter how many versions you end up creating, that the most important grade is always copied back to the default version.

### To copy any version quickly to the default version:

- 1 Choose Color > Memories > Save Memory A (Option-1). You can use any memory you want, but Memory A is convenient for this example.
- 2 Choose Color > Grade Version > Default (Command-U).
- 3 Choose Color > Memories > Load Memory A (Command-I) to apply the saved memory to the default version.

### To jump immediately to the default version, do one of the following:

- Choose Color > Grade Version > Default (Command-U).
- Press DEFAULT VERSION on the T-bar panel.

## Deleting Unused Versions

In cases where you want to make absolutely sure there's no alternative to the grade you want a clip to have, there's an easy way to delete all versions except for the version that's currently in use, thereby making it the default version.

### To eliminate either local or remote versions except for the current version:

- 1 Choose the local or remote version you want a clip to use.
- 2 Right-click the thumbnail of the clip you want to eliminate extra versions from, and choose Delete Unused Versions from the contextual menu.

If you selected a local version, all unused local versions will be deleted, but remote versions will be left alone. Similarly, if you selected a remote version, all unused remote versions will be deleted, but local versions will be left alone.

### To eliminate all other local and remote versions except for the current version:

- 1 Choose the version you want a clip to use.
- 2 Right-click the thumbnail of the clip you want to eliminate extra versions from, and choose Delete All Versions from the contextual menu.

Regardless of whether you selected a local or remote version, all other local and remote versions not in use will be deleted.

## Rendering Versions

When the time comes to render your clips in the Deliver page, each clip's currently selected version will be rendered. If you need to render a different version for a given clip, you can either make sure it's selected in the Color Page timeline before you open the Deliver page, or use the Versions submenu in the Color mode of the Deliver page Thumbnail timeline. This contextual menu also provides access to the Stereo 3D commands, the Edit PAR (Pixel Aspect Ratio) command, and a command for updating the Render Window Timeline thumbnails to reflect any changes you've made in case they haven't updated automatically.

Additionally, the Commercial Workflow output option, located in the Deliver page, provides a method of rendering multiple versions for each clip when outputting your project in Source Order (as individual media files). There are two additional options in the Version submenus of each clip in the Thumbnail timeline contextual menu that let you control which versions are rendered when you use Commercial Workflow.

**Render Disabled:** Turning this option on excludes that version from being rendered when Commercial Workflow is enabled.

**Enable Flat Pass:** Turning this option on forces the selected version to render with the grade turned off, essentially outputting the original media.

For more information on rendering versions using the Commercial Workflow option, see *Chapter 187, "Rendering Media."*

# Copying Grades

There are various methods you can use to copy grades from one clip to another. Which is appropriate to your need depends on your style of working with DaVinci Resolve.

## Protecting Adjustments with the Copy Grade Options

Before going into the myriad of ways that grades can be copied from one clip to another, you should know of a series of options, available from the contextual menu of the Gallery (right-click anywhere in the gray area of the Gallery), that let you carefully specify grading, sizing, and stereo data to be preserved when overwriting grades in clips you're copying to. There are three options:

**Copy Grade: Preserve number of nodes:** Lets you choose 0–10 nodes to be protected when applying a grade. When set to 1, the first node of the copied grade is ignored, but all other nodes are copied. When set to 5, the first five nodes of a copied grade are ignored, as long as there are at least five nodes in the grade of the clip you're copying to. This option is useful for colorists who routinely use the first few nodes for shot matching and scene balancing, with additional nodes applying individual or stylistic adjustments.

**Copy Grade: Preserve Camera Raw Settings:** When enabled, the camera raw Source settings of the current clip are preserved, letting you apply stylistic grades from unrelated clips without overwriting clip-specific source settings.

**Copy Grade: Preserve Input Sizing:** When enabled, Input Sizing adjustments are not overwritten by those of the copied grade.

**Copy Grade: Preserve Convergence:** When enabled, Convergence adjustments in the Stereo 3D palette are not overwritten by those of the copied grade.

**Copy Grade: Preserve Floating Windows:** When enabled, Floating Windows adjustments in the Stereo 3D palette are not overwritten by those of the copied grade.

**Copy Grade: Preserve Auto Align:** When enabled, Auto Alignment adjustments in the Stereo 3D palette are not overwritten by those of the copied grade.

**Copy Grade: Preserve Dolby Vision™ Analysis Metadata:** When enabled, Dolby Vision Analysis Metadata is not overwritten by that of the copied grade. Only available if you're set up to do Dolby Vision grading.

For more information, see *Chapter 9, "Data Levels, Color Management, and ACES."*

**Copy Grade: Preserve Dolby Vision™ Trim Metadata:** When enabled, Dolby Vision Trim Metadata in the Dolby Vision palette is not overwritten by that of the copied grade. Only available if you're set up to do Dolby Vision grading.

For more information, see *Chapter 9, "Data Levels, Color Management, and ACES."*

## Copy and Paste From One Clip to Another

The simplest way to copy color adjustments from one clip and apply them to another clip is to use the same Edit > Copy and Paste commands shared with nearly every other application ever made. However, you can use this straightforward functionality in one of two ways, depending on the focus of the interface.

- If you click on the clip in the Thumbnail Timeline, you copy and paste the entire grade from one clip to another.
- If you click inside the Node Editor, you copy and paste only the selected node to another selected node.

You can override this focus-based selection, and make the default behavior to only copy between nodes rather than entire grades regardless of the interface focus. Do this by checking the “Always perform copy and paste on selected nodes” box in the Color section of the User panel in Preferences.

### Methods of copying and pasting grades and node adjustments.

- **To copy and paste an entire grade:** Choose a clip with a grade you want to copy, and press Command-C. Then select the clip you want to copy the grade to in the Thumbnail Timeline, and choose Command-V to paste the entire grade.
- **To copy and paste a single node:** Choose a clip with a node you want to copy, and press Command-C. Then select the clip you want to copy the node to in the Thumbnail Timeline, and then click anywhere inside the Node Editor. Then choose Command-V to paste the node that was selected when you copied to the currently selected node in the node tree.

## Copy and Paste Specific Parameters Using Paste Value

The Edit > Paste Value (Shift-Option-V) command lets you paste only the value of a specific parameter into the node of another clip.

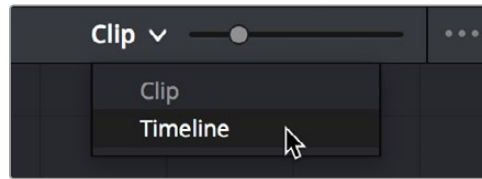
### To paste only a specific value:

- 1 Select a Corrector node with a parameter you want to copy, and press Command-C to copy it. You’re really copying the entire grade, but that’s okay.
- 2 Choose any other node in any grade, and open the palette that contains the parameter you want to paste into.
- 3 Double-click the number field of the specific parameter you want to paste the copied value into.
- 4 Choose Edit > Paste Value (Shift-Option-V) to paste that value.

**NOTE:** Paste Value only works for parameters within palettes; it does not work for OFX or Resolve FX in the Inspector.

## Applying Saved Grades From the Gallery

Every time you save a still into the Gallery or a memory, it contains both the Clip grade, and the Timeline grade, if one is applied. Which grade is applied from a still using the Apply Grade or Append Node Graph commands depends on which mode the Node Editor is in. This is selectable from the drop-down menu in the upper right-hand corner of the Node Editor.



Node Editor Clip and Track selector

If you're in Clip mode, you'll only copy the saved clip grade. If you're in Timeline mode, you'll only copy the saved timeline grade.

**NOTE:** This mechanism also works for saving and applying grades for clips that are in Groups. The currently selected group grade in the Node Editor determines which grade is saved with a still, and the currently selected mode in the Node Editor determines where a saved grade will be applied.

## Preparing to Copy or Apply Grades

For all other methods of copying or applying grades described in this chapter, there are some common methods of controlling how adjustments from the clip or still you're copying from are applied to the clip you're pasting or applying to.

### Choosing Which Aspect of a Grade to Apply with All/Color/Sizing

When applying grades using any of the techniques described in this section, you can use the All/Color/Sizing setting that are available via the Mark > Keyframe Timeline Mode submenu, the drop-down menu that's visible at the upper-right of the Keyframes Editor. When copying a grade in conjunction with the settings in this menu, the following rules apply:

- **All:** When All is selected, both the grade and the sizing are copied.
- **Color:** Only the grade is copied, the target clip retains its original Input Sizing settings.
- **Sizing:** Only the Input Sizing is copied, the target clip retains its original grade.

### Choosing How to Copy Keyframes

When copying or applying grades that have keyframes, you can choose how these keyframes are copied via a setting in the Gallery contextual menu by right-clicking anywhere in the gray background area of the Gallery, and choosing one of these options from the Apply Grades Using submenu. There are three options:

- **No Keyframes:** No keyframes are copied. The state of the grade at the frame used to save the still is what is applied to the target clip or clips.
- **Keyframes Aligning Source Timecode:** Keyframes are copied aligning the source timecode of the saved grade with the source timecode of the target clip. This is the ideal setting when you're copying a grade back to the clip it came from originally, or to a duplicate of that clip elsewhere in the Timeline, and you want the keyframes to align with the same frames as before. If there is no source timecode overlap, keyframes will be pasted aligning with the start frame of the edit, the same as the third option (below).
- **Keyframes Aligning Start Frame:** Keyframes are copied aligning the start frame of the clip that still was saved from with the start frame of the target clip. This is the ideal setting when you're copying a grade with keyframes from one clip to a completely different clip, with different timecode.



From that point on, keyframes, should they exist, will be copied or applied using the selected method whenever you copy a grade using any of the previously described methods.

## Copying Grades Using the Pointer

An incredibly easy way of copying a grade from one clip to another, or from a still or memory in the Gallery to a clip, is by using the third button of your mouse, usually mapped as the “middle-click” of the mouse scroll wheel.

### To copy a grade from a clip or still to one or more clips using the pointer:

- 1 Select the clip thumbnail you want to copy the grade to in the Timeline; a single selected clip appears highlighted orange. If you want to copy a grade to several clips, you can either Command-click multiple non-adjacent clips, or Shift-click a continuous range of clips; multiple selected clips appear highlighted red.
- 2 Middle-click the clip thumbnail, Gallery still, or memory you want to copy the grade from.

The grade of the clip you middle-clicked is copied to the previously selected clip or clips.

**TIP:** If you're using a trackpad, third-party software may allow you to define a “three-finger-click” that works the same as mouse button 3. If you're a pen and tablet or trackball user, you may be able to define a stylus or other button to perform the same button-3 action.

If you don't have access to a middle-click on the pointing device you're using, there's another way of doing this using a contextual menu command.

### To copy a grade from a clip or still to one or more clips using the Apply Grade command:

- 1 Select the clip thumbnail you want to copy the grade to in the Timeline; a single selected clip appears highlighted orange. If you want to copy a grade to several clips, you can either Command-click multiple non-adjacent clips, or Shift-click a continuous range of clips; multiple selected clips appear highlighted red.
- 2 Right-click the clip thumbnail, Gallery still, or memory you want to copy the grade from, and choose Apply Grade from the contextual menu.

The grade of the clip you middle-clicked is copied to the previously selected clip or clips.

## Copy Forward Commands

Another simple way of copying grades is to use the equals (=) and minus (–) keys on the keyboard to copy grades forward from one or two clips behind the currently selected clip. This is a great way to copy grades in scenes with a shot-reverse-shot structure, where you're cutting between two angles of coverage, each of which uses the same grade.

### To copy grades forward:

- **To copy a grade from one clip back:** Choose Color > Apply Grade From One Clip Prior, or press Equals (=).
- **To copy a grade from two clips back:** Choose Color > Apply Grade From Two Clips Prior, or press Minus (–).

## Copying Using Memories

Memories are virtually identical to stills, except that they're labeled with a letter (A–Z) for easy access via keyboard shortcuts. A memory bank above the Gallery browser provides a visual reference of which saved grade is assigned to which letter. This makes it easy to keep track of multiple saved memories in instances where you're copying them to several different clips throughout a scene or program.

For example, you might save a memory for each angle of coverage in a complicated scene, making it easy to copy grades forward. In another example, you might save a memory for the grade applied to each interview subject's headshot in a documentary, so you can copy that grade forward as you work through the documentary.

**To save the current clip's grade to a memory for future use, do one of the following:**

- Choose Color > Memories > Save Memory A–H (Option-1–8).

If you save a grade to a memory that already contains something, the previous memory is overwritten.

**To apply a memory to one or more clips in a timeline, do one of the following:**

- 1 Choose one or more clips in the Thumbnail timeline to copy to. This can be the current clip, or it can be a range of clips that you select by Command-Clicking or Shift-Clicking.
- 2 Do one of the following to copy a grade to the selected clips:
  - Right-click a memory and choose Apply Grade.
  - Choose Color > Memories > Load Memory A–H (Command-1–8).

**To clear a memory:**

- Right-click a memory, then choose Clear.

## Copying Using Preview Memory

You can also preview the effect of a memory or saved grade on the current clip, with the option to either keep it if it works, or go back to the previous grade if it doesn't.

**To Preview Memory:**

- 1 Move the playhead to a clip for which you want to preview a memory.
- 2 Choose Color > Preview Memory (Option-Shift-P).
- 3 Do one of the following:
  - Right-click any saved still in the Gallery and choose Apply Grade.
  - Choose Memories > Load > Memory A–H (Command-1–8).

The selected grade or memory is now being previewed.
- 4 Now, do one of the following to either accept or reject the memory:
  - If you like the effect, then you can leave it be and move on to another clip.
  - If you don't like the effect, then choosing Color > Preview Memory (Option-Shift-P) again reverts the clip to the original grade.

## Copying from Stills in the Gallery

As mentioned previously, each still you store in the Gallery contains the grading information as well as the image of the frame it was saved from. This saved grade can be applied to any clip in the Timeline.

**IMPORTANT:** Copied grades overwrite any previously existing grades on the clip or clips you copy to.

### To apply a grade from a still to one or more selected clips in a timeline:

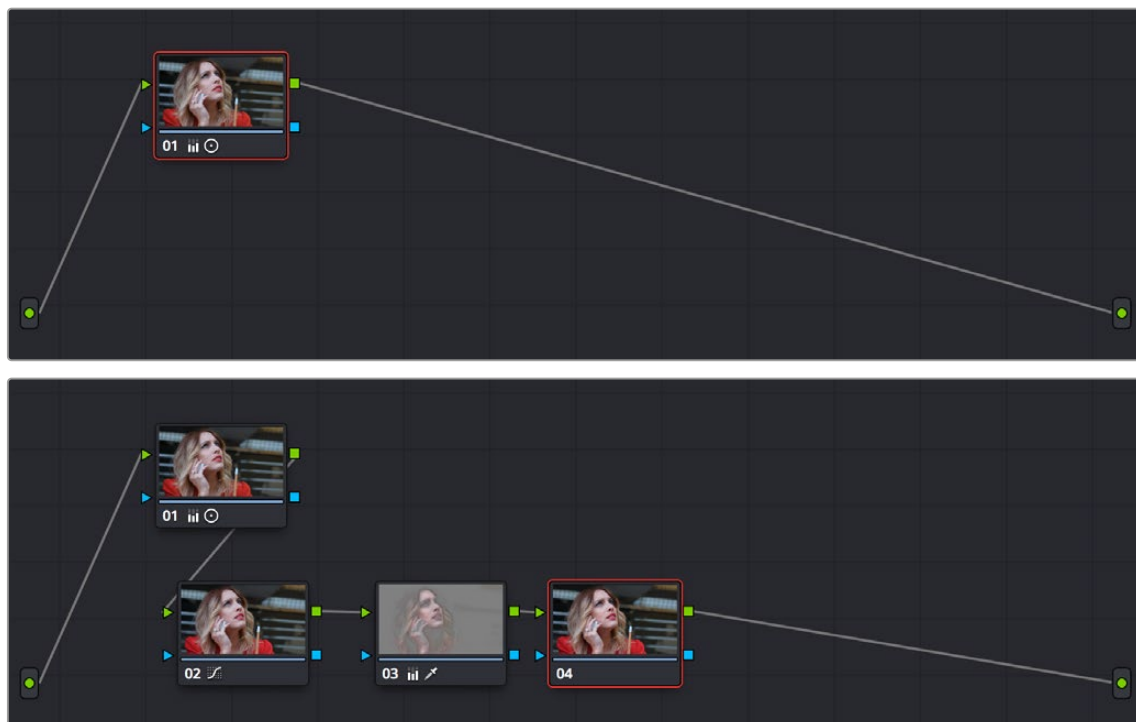
- 1 Choose one or more clips in the Thumbnail timeline to copy to. This can be the current clip, or it can be a range of clips that you select by Command-clicking or Shift-clicking.
- 2 Do one of the following to copy a grade to the selected clips:
  - Drag the still from the Gallery into the Viewer.
  - Right-click a still in the Gallery and choose Apply Grade.
  - Middle-click a still in the Gallery.
  - Double-click a still to wipe it in the Viewer against the current clip in the Timeline, then right-click the Viewer and choose one of the options from the Apply Grade submenu.

## Append Node Graph

You can also append a saved grade to any clip in a timeline. This adds the entire node graph of the saved still to the end of the node tree of the current clip. In other words, the current clip's grade isn't overwritten, the applied grade is added to the end of it.

By planning ahead, you can save fragmentary grades that create specific effects or adjustments using only a few nodes. You can then use these fragmentary grades as a toolkit that you can add to other grades in order to mix and match different adjustments and effects.

For example, you could create a three-node glow effect, save it, and then apply that effect at the end of a completely different clip's grade.



Appending grades places them after the original nodes

**To append a saved grade or memory as individual nodes, do one of the following:**

- Drag a still or memory from the Gallery onto a connection line in the Node Editor; when the plus icon appears, drop it and its nodes will be appended within the node tree starting at that connection.
- Right-click a still or memory in the Gallery, and choose Append Node Graph.

Ordinarily, when you append the node graph from a memory or still to another node graph, you end up adding a whole shedload of new nodes. This may be exactly what you need, but in situations where you want to keep things a bit tidier, you also have the option of appending node graphs as compound nodes.

**To append a saved grade or memory as a compound node:**

- Command-drag a still or memory from the Gallery onto a connection line in the Node Editor; when the plus icon appears, drop it and its nodes will be appended to the node tree as a single compound node.

**NOTE:** When you append the nodes from a gallery still to a grade, how keyframes are applied depends on the “Apply Grades Using” setting of the Gallery.

For more information, see *Chapter 141, “Using the Gallery.”*

## Aligning Keyframes to a Specific Frame While Copying Grades

If you need to copy a grade with keyframes so that the start keyframe of the copied grade aligns with a specific frame of the Timeline, you can do so using the following procedure:

**To copy a grade and align its keyframes to a specific frame of the Timeline:**

- 1 Save a grade with keyframes as a Gallery still by right-clicking the Viewer and choosing Grab Still.
- 2 Choose the clip you want to copy the saved grade to in the Thumbnail timeline.
- 3 Double-click the Gallery still to wipe it against the current clip in the Viewer.
- 4 Move the playhead to the frame of this clip you want the first keyframe of the saved grade to be aligned with.
- 5 Right-click the Viewer and choose Apply Grade > Align Keyframes to Current Frame.

## Copying Individual Nodes and Settings

Copying grades from clip to clip copies everything except for keyframes and motion tracking. However, there is one way you can copy the motion tracking from one clip to another, and that’s by copying and pasting individual node settings. This can save time when you’re assembling complicated node trees by recycling specific nodes or node settings from previous grades.

### Copying and Pasting All Settings From One Node to Another

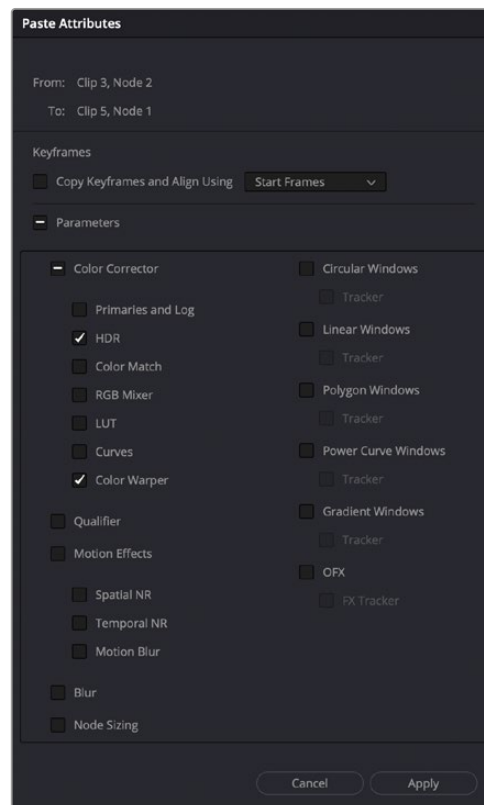
The simplest thing you can do is to copy all of a node’s settings, and paste them into another node. This makes it easy to duplicate things like windows, qualifier settings, keyframing, or motion tracking that you want to reuse in another node as the basis for another operation. This is also a quick way to manually ripple a change you make in a node to that same node in another clip’s grade.

**To copy a node's settings from one clip to another, do one of the following:**

- Option-drag one node onto another. When you drop it, the settings of the node you dragged overwrite those of the node you dropped onto.
- Select a node with settings you want to copy and choose Edit > Copy (Command-C). Then, select a node you want to paste these settings to either in the current grade or in the grade of another clip, or create a new node, and choose Edit > Paste (Command-V) to paste the settings you copied. These pasted node settings overwrite any other settings that node previously used.

## Paste Attributes on the Color Page

You can copy the settings of one node and paste a subset of those settings to another selected node by choosing Edit > Paste Attributes (Option-V). A Paste Attributes window appears that shows you which node you're copying from, and which node you're pasting to, and provides controls for choosing whether you want to paste keyframes and how to align them (Start Frames or Source Timecode), as well as a series of checkboxes for choosing which palettes, windows, and OFX/ResolveFX you want to apply. You can only paste all parameters from a specified palette; there's no current provision for pasting specific parameters from within a given palette, with the exception of pasting specific windows and their accompanying trackers.



The Paste Attributes window on the Color page

## Copying from the Node Graph of Other Clips or Gallery Stills

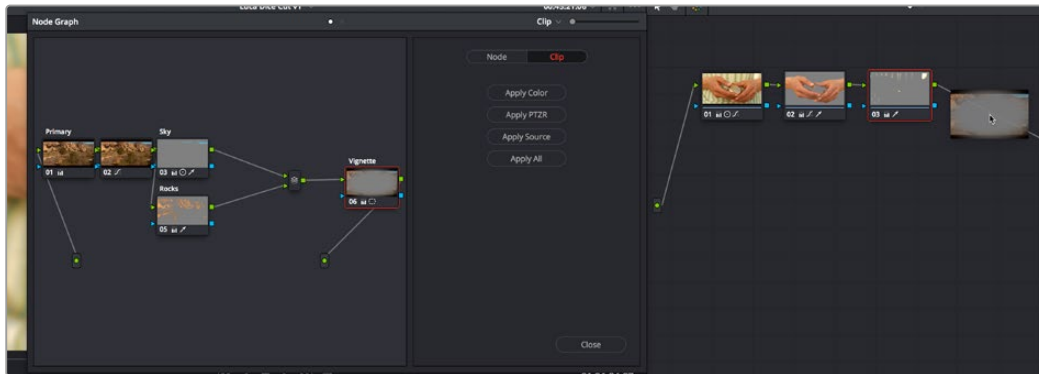
When constructing new grades, you may sometimes find it convenient to copy existing nodes or node settings from the grade of another clip or gallery still. This can be easily accomplished by exposing the node tree of any still that's saved in the Gallery, or any clip in the Thumbnail timeline, and using the controls found in the floating node graph to copy individual nodes or node adjustments to the current clip's grade shown in the Node Editor.

### Copying Clip Settings Using Display Node Graph

In Clip mode, the floating Node Graph window has four sets of controls you can use to choose how to copy nodes and adjustments:

- **Clip Node Graph:** By default, the Clip node graph is displayed at the left. You can drag any node from the floating node graph to the current grade shown in the Node Editor, and drop it onto an existing node to overwrite that node's settings or onto a connection line to insert it as a new node.

- **Timeline Node Graph:** (Only available if there is a Timeline grade) You can switch to the Timeline grade, if there is one, by either clicking the second button in the Node Graph title bar, or by choosing Timeline from the drop-down menu at the upper right-hand corner of the floating Node Graph window.
- **Apply Color/PTZR/Source/All buttons:** With the Clip panel selected, four buttons let you selectively copy the entire grade, the sizing, the source settings, or all settings at once to the current clip.

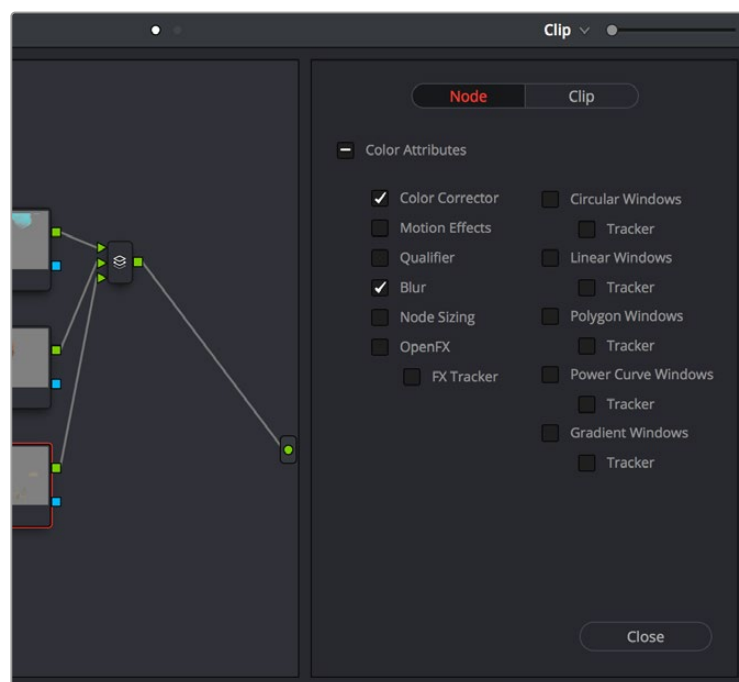


Dragging a node from another clip's node graph to the currently selected clip's grade in the Node Editor

## Copying Node Settings Using Display Node Graph

In Node mode, the floating Node Graph window displays all of the available Color Attributes found within each node, instead of the Apply Color/PTZR/Source/All buttons:

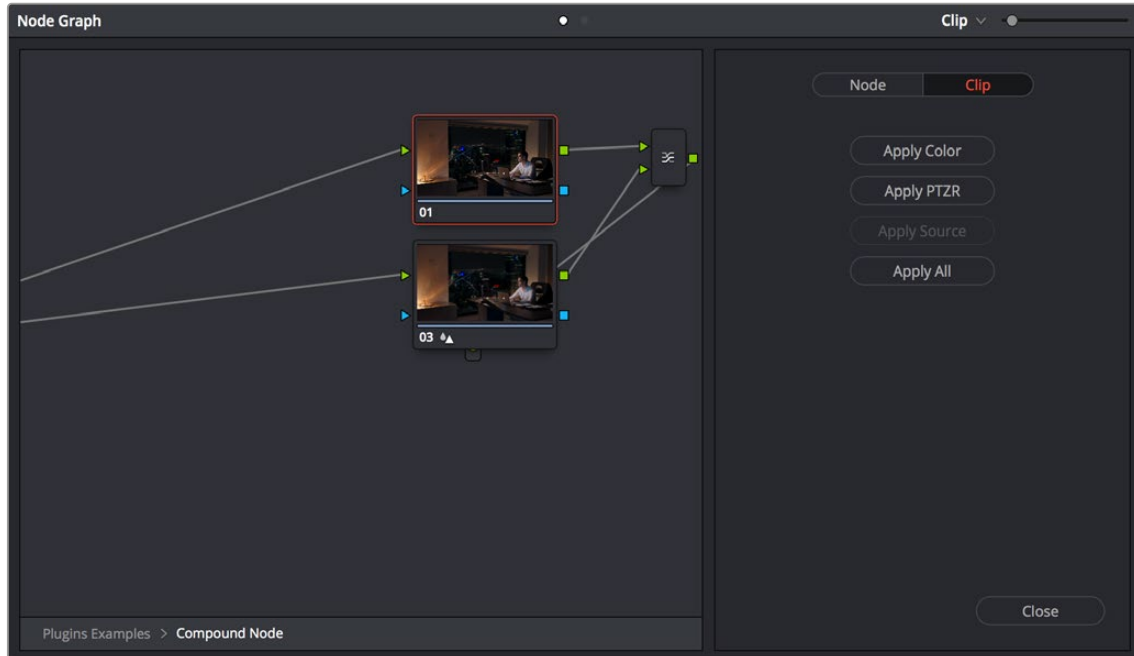
- **Node Settings:** Use the checkboxes to choose which node adjustments get copied and which do not. After you've checked the settings you want to selectively copy, dragging a node from the floating node graph onto nodes in the Node Editor copies or appends the selected attributes into the target node, leaving all other attributes alone. For Windows and OpenFX, you have the option of also copying any motion tracking that's available.



The Node panel open with the Color Corrector and Blur attributes selected for copying

## Ability to Open Compound Nodes in “Display Node Graph”

When you right-click a Gallery still or a thumbnail and choose Display Node Graph for a grade that uses compound nodes, you can right-click any compound node and choose “Show compound node,” or Command-double-click a compound node to open it and see its individual nodes.



Opening a compound node in a floating Node Graph window

## How to Copy Nodes Using Display Node Graph

The following procedures describe how to open the floating node graph in different situations.

### To copy individual nodes or settings from any still in the Gallery:

- 1 Click the thumbnail in the Timeline of the clip you want to copy nodes to. Its node graph will appear in the Node Editor.
- 2 Right-click a still in the Gallery you want to copy nodes from, and choose Display Node Graph. A floating Node Graph window appears displaying that still's node tree.
- 3 (Optional) While the Gallery Node Graph window is open, you can select any other still, and this window will update to show the currently selected still's node graph, ready for you to copy from.
- 4 In the floating Node Graph window, choose Clip if you want to copy overall clip attributes, or choose Node to copy individual node attributes.
- 5 Do one of the following to copy nodes or settings to the Node Editor.
  - In Clip mode, drag any node from the floating node graph to the current grade shown in the Node Editor, and drop it onto an existing node to overwrite that node's settings. If you open the Node panel and choose specific attributes, only those attributes will be copied to the node you drag onto.
  - In Clip mode, drag any node from the floating node graph to the current grade shown in the Node Editor, and drop it onto a connection line to insert it as a new node. If you open the Node panel and choose specific attributes, only those attributes will be copied to the new node that's created.
  - In Clip mode, click Apply Color to copy the entire grade from the floating node graph to overwrite the current grade in the Node Editor.

- In Clip mode, click Apply PTZR to copy the sizing from the floating node graph to overwrite that of the current clip.
  - In Clip mode, click Apply Source to copy the source settings from the floating node graph to overwrite those of the current clip.
  - In Clip mode, click Apply All to copy every setting from the floating node graph to the current clip.
  - In Node mode, select which attributes you want to copy, and then drag any node from the floating node graph to the current grade shown in the Node Editor, and drop it onto an existing node to overwrite the settings of that node that you selected.
  - In Node mode, select which attributes you want to copy, and then drag any node from the floating node graph to the current grade shown in the Node Editor, and drop it onto a connection line to insert it as a new node with only the settings that you selected.
- 6 While the floating Node Graph window is open, you can also select any clip in the Thumbnail timeline to change the grade that's displayed in the Node Editor, ready for you to copy to.
  - 7 Click Close when you're finished.

**To copy individual nodes or settings from any clip in the Timeline:**

- 1 Click the thumbnail in the Timeline of the clip you want to copy nodes to. Its node graph will appear in the Node Editor.
- 2 Right-click the thumbnail of another clip you want to copy nodes from, and choose Display Node Graph.

**NOTE:** The Display Node Graph command only appears in the contextual menu of a clip in the Thumbnail timeline that is not currently selected.

- 3 (Optional) In the floating Node Graph window, choose Clip if you want to copy overall clip attributes, or choose Node to copy individual node attributes.
- 4 Do one of the following to copy nodes or settings to the Node Editor.
  - In Clip mode, drag any node from the floating node graph to the current grade shown in the Node Editor, and drop it onto an existing node to overwrite that node's settings. If you open the Node panel and choose specific attributes, only those attributes will be copied to the node you drag onto.
  - In Clip mode, drag any node from the floating node graph to the current grade shown in the Node Editor, and drop it onto a connection line to insert it as a new node. If you open the Node panel and choose specific attributes, only those attributes will be copied to the new node that's created.
  - In Clip mode, click Apply Color to copy the entire grade from the floating node graph to overwrite the current grade in the Node Editor.
  - In Clip mode, click Apply PTZR to copy the sizing from the floating node graph to overwrite that of the current clip.
  - In Clip mode, click Apply Source to copy the source settings from the floating node graph to overwrite those of the current clip.
  - In Clip mode, click Apply All to copy every setting from the floating node graph to the current clip.
  - In Node mode, select which attributes you want to copy, and then drag any node from the floating node graph to the current grade shown in the Node Editor, and drop it onto an existing node to overwrite the settings of that node that you selected.
  - In Node mode, select which attributes you want to copy, and then drag any node from the floating node graph to the current grade shown in the Node Editor, and drop it onto a connection line to insert it as a new node with only the settings that you selected.



- 5 (Optional) While the floating Node Graph window is open, you can also select any clip in the Thumbnail timeline to change the grade that's displayed in the Node Editor, ready for you to copy to.
- 6 Click Close when you're finished.

## Rippling Adjustments Among Multiple Clips

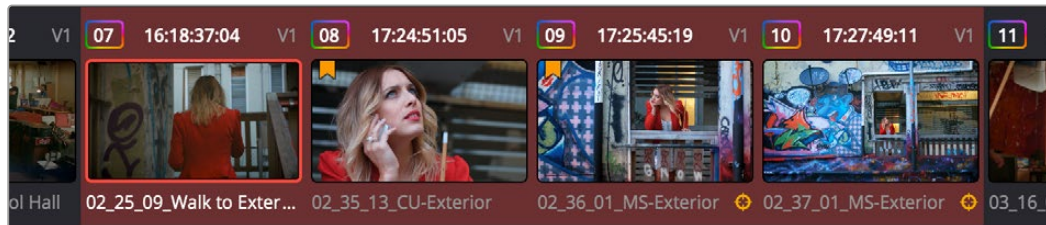
If you've graded a sequence of clips and find yourself needing to make a quick change to several clips all at once, you can use the Color > Ripple Node Changes to Selected Clips/Current Group commands to quickly copy changes made to one clip to several others. However, these commands require that you follow certain rules in order to get the results you want.

Node rippling uses the number of each node to determine what changes should ripple to which nodes. In other words, changes made to Node 3 will ripple to Node 3 of every other selected clip, or to Node 3 of every other clip in the current group (depending on which command you use).

Rippling changes from one clip to several others requires that all clips share the same number of nodes, at least for the nodes you're rippling. If you try to ripple a change to one or more clips that don't have a node with the same number as the one you're rippling, then no change will be rippled to those clips.

### To ripple a change in one node to several selected clips:

- 1 Select the clip you want to use to make the change so that it's the current clip, outlined in orange.
- 2 Now, Command-click or Shift-click to select the range of clips you want to ripple your change to, so that they're all selected red.



The orange outlined clip is the current clip you're modifying, and the clips highlighted in red are the ones you're rippling this change to

You don't have to select the clips you're going to ripple your change to prior to making the change, but it may be easier to keep track of what you're doing if you set everything up in advance. The clip you're changing should be outlined in orange, and the clips you're rippling the change to should be highlighted in red.

- 3 Select a node to modify. Ideally, every selected clip should have a node sharing the same number as the one you're modifying. If some clips don't, then those clips won't inherit the change you're going to be rippling.
- 4 Make whatever changes you need to, adjusting any of the palette controls in the Color page you need to, except for Camera Raw, Sizing, and Data Burn, which aren't node-specific.
- 5 When you're happy with the change you've made, choose Color > Ripple Node Changes to Selected Clips.

The change you've made should ripple to every other selected clip with a node of the same number, and in a few moments the thumbnails of those clips should update to show the change.

If you've created a group, it's even easier to ripple node adjustments to the other clips in the group.

#### To ripple a change in one node to the current group:

- 1 Select the clip you want to use to make the change so that it's the current clip, outlined in orange. It must be part of a group for this to work.
- 2 Select a node to modify. Ideally, every selected clip should have a node sharing the same number as the one you're modifying. If some clips don't, then those clips won't inherit the change you're going to be rippling.
- 3 Make whatever changes you need to, adjusting any of the palette controls in the Color page you need to, except for Camera Raw, Sizing, and Data Burn, which aren't node-specific.
- 4 When you're happy with the change you've made, choose Color > Ripple Node Changes to Current Group.

The change you've made should ripple to every other clip in the current group that has a node of the same number, and in a few moments the thumbnails of those clips should update to show the change.

## Appending a Node to Multiple Clips

If you've made an adjustment to a node in one clip that you'd like to apply to several other clips, you can quickly do that using the Color > Append Node to Selected Clips command. You can do this as a prelude to using one of the Ripple Node Change commands, as it's a quick way to make sure several previously ungraded clips have the same nodes, but you can also do this as a quick way to apply the same node change to several clips, regardless of whether they have matching node trees.

#### To append a node to other selected clips:

- 1 Select the clip you want to use to make the change so that it's the current clip, outlined in orange.
- 2 Select the node you want to append to other clips.
- 3 Now, Command-click or Shift-click to select the range of clips you want to ripple your change to, so that they're all selected red.

The clip you're copying from should be outlined in orange, and the clips you're rippling the change to should be highlighted in red.

- 4 Choose Color > Append Node to Selected Clips.

The selected node should be appended to the end of the node tree of each selected clip, and in a few moments the thumbnails of those clips should update to show the change.

## Using Shared Nodes

Shared nodes are meant to be a way to extend the benefits of automatically rippled changes among different clips to colorists that prefer a flatter node structure than Group Grading allows. By turning individual Corrector nodes into Shared nodes, and copying these to multiple clips, you enable linked adjustments right from within the clip grade. This means that the clip grade can freely mix both clip-specific nodes and Shared nodes, all within the same node tree. This makes Shared nodes fast to use as there's no need to create groups or switch to a group node tree (covered in the next section) to reap the benefits of linked adjustments among multiple clips.



A grade with an Unshared (at left) and Shared node (at center), a badge indicates the Shared node, which is also locked

## What Are Shared Nodes Good For?

Shared nodes are similar to group grades, except that they don't require grouping and can be added to any normal grade. Changes made to a Shared node are automatically rippled to all other instances of that node in the grades of other clips. Furthermore, you can add as many Shared nodes to a grade as you like, and you can arrange them in any order to control the order of the operations they apply. And, of course, you can intersperse them with ordinary Corrector nodes

**Shared nodes are extremely flexible. For example, you can use Shared nodes to:**

- Add a Color Space Transform Resolve FX or a LUT to the beginning of every clip from a particular source
- Add a base correction to every talking head shot of a particular interviewee
- Add a shot matching adjustment to each clip from a particular angle of coverage within a scene
- Add a stylistic adjustment to every clip in a specific scene
- Make your base adjustments when grading with remote versions, so those adjustments remain linked when you copy your remote versions to local versions for fine tuning

In fact, you can mix and match Shared nodes among differently overlapping sets of clips to accomplish any or all of the above at once. For example, you can add one Shared node to make an adjustment to every clip from a particular camera, add a second Shared node to each of those clips that are in a particular scene, and then add a third Shared node to whichever of those clips happen to be a closeup of the lead actress, before adding one or two regular Corrector nodes that aren't shared to make clip-specific tweaks.

## Creating Shared Nodes

Creating a Shared node is easy, assuming you've created a node that has an adjustment you'd like to share among multiple clips.

**To create a Shared node:**

- Right-click any Corrector node and choose Save as Shared node.

## Locking Shared Nodes

Once you turn a node into a Shared node, that node is automatically locked, preventing you from accidentally making adjustments to it that would affect all other grades using that same Shared node.

**To toggle the locked status of a Shared node, do one of the following:**

- Right-click any Shared node and choose Lock Node from the contextual menu.
- Open the Keyframe Editor, and click the Lock icon in the track header of that node's keyframe track.

## Copying Shared Nodes

Because Shared nodes are essentially Corrector nodes within clip grades, they're easy to work with. Once you've created one or more Shared nodes, there are a variety of ways you can copy them to the grades of other clips in your program to take advantage of the linked adjustments they let you make.

### Ways of copying Shared nodes among multiple clips:

- **Add a Shared node to another clip's grade using the Node Editor contextual menu:** Once you save a node as a Shared node, it becomes available from the bottom of the Add Node submenu of the Node Editor contextual menu, making it easy to add any Shared node to any clip. If you customize the label of the Shared node, that custom label appears in the contextual menu, making it easier to find what you're looking for.
- **Add Shared nodes to a basic grade you'll be copying to other clips:** If you create one or more Shared nodes when you initially build a grade, copying that grade to other clips naturally copies the Shared nodes as well.
- **Save a Shared node as a Gallery still and apply it to other clips:** If you save a grade with a Shared node in it to the Gallery, then every time you copy that Gallery still to another clip, you copy its Shared node.
- **Create a Shared node and append it to a selection of additional clips:** If you've already graded several clips in a scene, you can add a Shared node to the end of one of the clips grades and make sure it's selected, then select all of the other clips in the scene and choose Append Node to Selected Clips.
- **Use Shared nodes to preserve linked adjustments when copying remote grades to local grades:** If you use Shared nodes to make your base adjustments when you grade using Remote Versions to automatically copy those grades to other clips that come from the same source media, those adjustments will remain linked when you copy your remote versions to local versions for fine tuning.

You can also choose options in the Gallery that dictate what happens when you copy a grade that has shared nodes within it.

## Converting Shared Nodes Back to Corrector Nodes

Sometimes you need to stop a node from being shared. For example, if you want to copy a grade from another scene to use as the starting point for a new grade, chances are you don't want any shared nodes to continue being shared as you customize that grade for the new scene. In this instance, you can convert the Shared node back to a regular Corrector node, make any adjustments you need to customize it for the new scene, and then turn that node into a brand new Shared node that's specific to the new scene.

### To convert a Shared node back to a Corrector node:

- Right-click any Shared node, and choose Convert to Corrector from the contextual menu.

## Deleting Shared Nodes

If you've created a Shared node that's being used in multiple clips, and you decide you want to eliminate the linked relationship among these nodes so they all work independently, you can "delete" a specific Shared node. This leaves the now unlinked nodes intact within each node tree in which they appear, but clears their effect. Additionally, that Shared node is removed from the Add Node submenu.

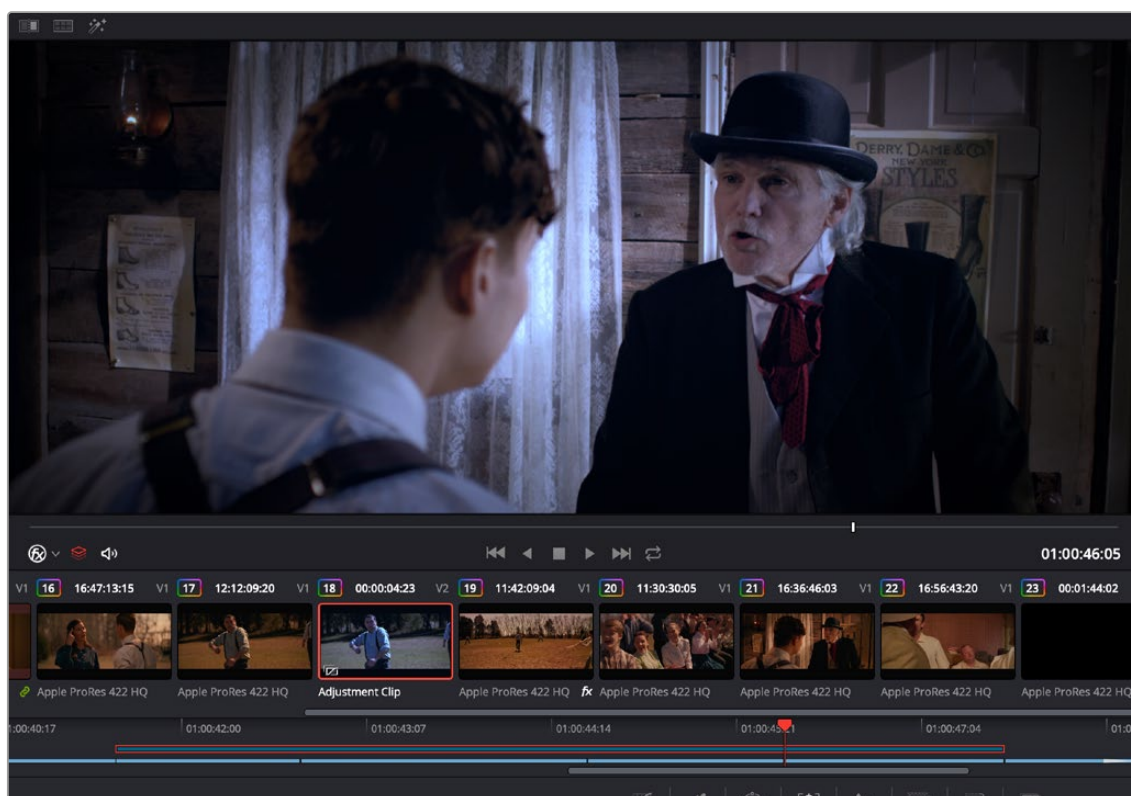
### To Delete a Shared node:

- Right-click any Shared node, and choose a node to delete from the Delete Shared Node submenu.

# Using Adjustment Clips

You can also apply a single grade to multiple clips in the Timeline using Adjustment clips that have been superimposed over numerous other clips in the Timeline. They can be used to apply a single grade to all clips in a scene, or to add further color adjustments or trims to a section of clips via a second grade that's applied on top of the individual clip grades that are already applied.

Adjustment clips are a great way to add adjustments to multiple clips that you may be making frequent changes to. If you make changes to a grade in an Adjustment clip, that change is automatically applied to all clips underneath it in the Timeline. Additionally, grades applied via adjustment layers can be turned off by disabling the adjustment layers, either one at a time, or by disabling an entire track with adjustment layers in it.



An Adjustment clip (shown selected in the Mini Timeline) used to apply a cooler grade to four other clips in a scene

Adjustment clips can be edited into the Timeline on the Edit page, and are available from the Effects bin of the toolbox in the Effects Library. When an Adjustment clip is superimposed above one or more clips in the Timeline, all effects that are applied to the Adjustment clip are also applied to all clips underneath it.

Adjustment clips can be used to apply the following types of effects:

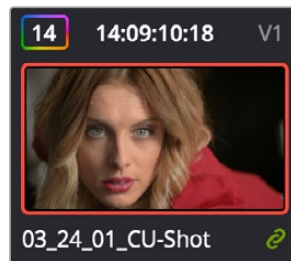
- ResolveFX and OpenFX plugins
- Inspector parameters, including Composite, Transform, Cropping, and Dynamic Zoom
- Fusion page effects
- Color page grading and sizing

Effects clips are a fast and easily revised way to apply one or more grades and effects to a range of clips.

**TIP:** Multiple adjustment layers can be grouped together so that you can apply Group and Clip grades to adjustment layers, linking multiple adjustment layers together in situations where you want to apply the exact same trim or stylistic adjustment to multiple scenes in non-contiguous areas of the Timeline.

## Using Groups

Grouping is one of DaVinci Resolve's most powerful features for organizing grades amongst multiple clips. Groups are manually created collections of clips with automatic linking relationships that make it easy for you to control the rippling of grades among sets of clips that you define. When clips have been grouped together, selecting one member of the group reveals a group "linked" badge at the top-right of each clip's thumbnail belonging to the group.



The green link icon indicates this clip is in a group

You can create as many groups within a single project as you need. You can add or remove shots from the current group, and the current group is defined by the last shot you've selected (if it belongs to a group, that's the new current group).

Once grouped, you can choose from one of four Node Editor modes to grade the current clip. Which mode you choose determines whether that grade will automatically ripple to every other member of that group, or whether the grade will only affect that one clip. In this way, you can combine multiple node trees, some which will ripple, and one which won't, to create the final look of each clip in that group.

For example, you might group together every clip from the same angle of coverage in a scene. You might also group together every clip from a particular section of B-roll that will be graded with the same treatment. This way, whenever you make a change or otherwise update one member of the group using one of the rippled Node Editor modes, every clip in the group will be changed together.

This section discusses how to create and manage groups, how best to use them, and what issues you need to be aware of to use groups to best advantage.

## Creating and Managing Groups

The following procedures describe how to create, load, modify, and delete groups.

### To create a new group:

- 1 Select one or more clip thumbnails in the Timeline. You can select a contiguous range of clips by clicking one and then Shift-clicking another, or you can make a noncontiguous selection by Command-clicking individual clip thumbnails anywhere in the Timeline.
- 2 Right-click one of the selected clip thumbnails, and choose Add Into New Group.
- 3 Type a name into the field of the Group Name dialog, and click OK.

The new group is created, and the clips you had selected now appear with the “linked” group badge above the thumbnails.

### To load a different group, do one of the following:

- Move the playhead to another clip that’s a member of that group.
- Right-click any clip belonging to any group, and choose Group > “Name of group” > Load.

The selected group now becomes the current group. Using the Add Into Current Group command will add a clip to the newly loaded group.

### To add one or more clips to an existing group:

- 1 Load the group you want to add clips to.
- 2 Select one or more clip thumbnails in the Timeline. You can select a contiguous range of clips by clicking one and then Shift-clicking another, or you can make a noncontiguous selection by Command-clicking individual clip thumbnails anywhere in the Timeline.
- 3 Right-click one of the selected clip thumbnails, and choose Group > “Name of group” > Assign to Group.

The selected clips now appear with the “linked” group badge above the thumbnails.

### To remove one or more clips from an existing group:

- 1 Select one or more clip thumbnails in the Timeline that you want to remove from a group. You can select a contiguous range of clips by clicking one and then Shift-clicking another, or you can make a discontiguous selection by Command-clicking individual clip thumbnails anywhere in the Timeline.
- 2 Right-click one of the selected clip thumbnails, and choose Remove From Group.

### To rename a group:

- 1 Right-click any clip belonging to any group, and choose Group > “Name of group” > Rename.
- 2 Type a new name into the field of the Group Name dialog, and click OK.

The group is renamed. The new name appears as a new submenu of the Timeline contextual menu.

### To delete a group:

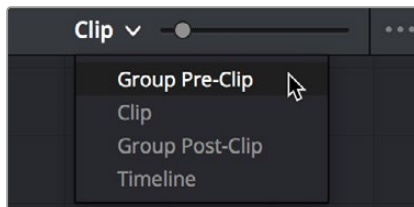
- Right-click a clip belonging the group you want to delete, and choose Group > “Name of group” > Delete.
- This clears the group, while preserving the last applied grade within each clip.

DaVinci Resolve makes a default smart filter for all clips in the current group. It can be found in the Clips drop-down, under Grouped > Current Group.



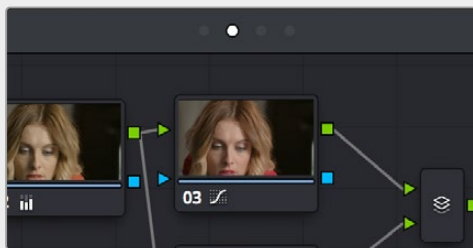
## Using Group Modes to Control Which Grades Ripple and Which Don't

When you create a group and then select any clip that is a member of that group, additional Node Editor modes become available that let you apply multiple levels of grades to the clips in that group.



Available options in the Node Editor modes drop-down

For clips that aren't grouped, two dots at the top of the Node Editor let you switch between Clip and Timeline modes via a single click. If you're working on a clip that's part of a group, four dots will be displayed to allow fast access to the Pre-Clip and Post-Clip Group modes as well.



Four dots at the top of the Node Editor let you choose which mode you're working on

Node trees you create using the Group Pre-Clip and Group Post-Clip modes automatically ripple changes to every member of a group. Node trees created with the Clip mode are specific to each clip. Node trees created with the Timeline grade affect the entire Timeline, all at once. Combining node trees made using these different modes makes it easy combine clip-specific changes with group-wide changes that make grading faster.

- **Group Pre-Clip:** Node trees assembled in this mode affect every clip in the group simultaneously. Pre-Clip adjustments are useful for creating an overall grade that you want to use as a starting point for the scene, such as when you need to normalize a group of log-encoded clips in a specific way, or when you want to create a base grade that generally improves every clip in a scene.
- **Clip:** The Clip grade corresponds to the local or remote version associated with that clip. Node trees assembled in the Clip mode only affect the specific clip that's selected. All Local and Remote versions you create are in Clip mode. Clip adjustments are useful for matching clips to one another in a scene, doing clip-specific secondary adjustments, or making any other sort of adjustment that you only want applied to the current clip, rather than to the group as a whole. Clip adjustments are applied after Pre-Clip adjustments.



- **Group Post-Clip:** Node trees assembled in this mode also affect every clip in the group simultaneously, but these adjustments are applied after the Clip adjustments. Post-Clip adjustments can be useful for applying a creative look to an overall scene, so that later revisions to a scene's look can be accomplished with a single set of adjustments that automatically affect the entire group.
- **Timeline:** Node trees assembled in this mode affect every single clip in the entire Timeline. Timeline adjustments are useful when you have a single change that you want to apply to every single clip at once.

The image processing of each Node Editor mode is sequential, making it easy to keep track of the order of operations affecting a particular clip.

## An Example of Grading a Group of Clips

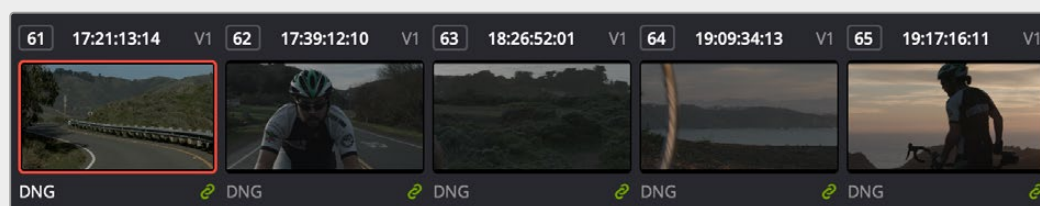
This section shows an example of how you might approach grading a group of clips using the different Node Editor modes that are available. Keep in mind that this is only one of many possible approaches you could take to the use of groups, and was chosen because it exercises everything you can do to a single grade.

To get started, select a series of clips that you want to group, and right-click one of them and choose Create New Group to group them together, entering a name and clicking OK when prompted. One convenient advantage of simply creating groups from a selection is that you're not limited to selecting a continuous range of clips in the Timeline. Another advantage is that you can use the View > Timeline Thumbnail Mode > Source (C-mode) command, or Timeline filtering, to isolate the range of clips you want to use for creating a group.

There are different ways you might choose to group clips together:

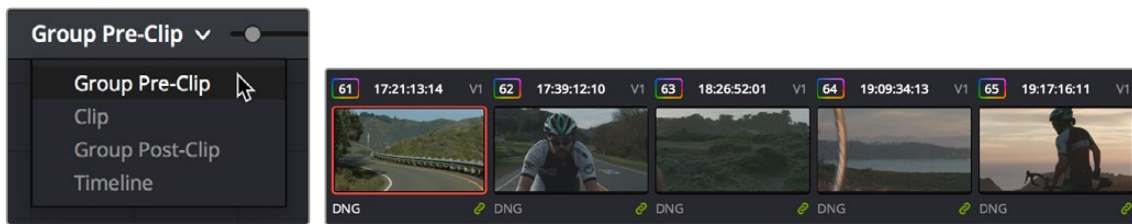
- All clips within a particular scene
- All clips recorded from a specific camera
- All headshots of a particular person
- All clips from a particular angle of coverage
- All exteriors of a particular location

The list could go on and on, but you get the idea. You can group any selection of clips together that have reason to share a common grade. In this example, a series of shots from an exterior scene are selected and made into a group.



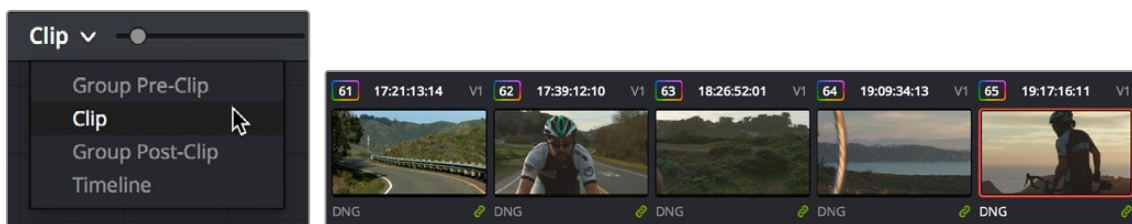
Clips made into a group

Once you've created a group, choose Group Pre-Clip from the Node Editor mode drop-down. This is the first grade that's applied, as the underlying grade that affects the entire group. Using this node tree, adding a LUT to normalize the clips, a Color Wheel adjustment, and some Curve adjustments to tune the color provide a primary grade that now affects every single clip in the group.



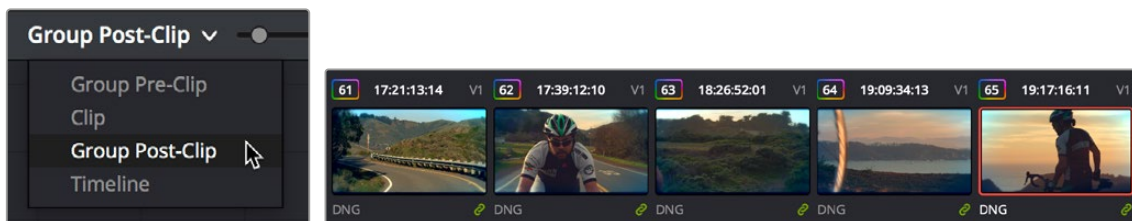
A primary adjustment applied to the Group Pre-Clip mode

With this accomplished, you can see there's some variation among the clips in the scene that needs to be corrected in order for every clip to match. Choose Clip from the Node Editor mode drop-down. Grades that you make in Clip mode only affect the clip to which they're applied, so now you can work clip by clip using the Image Wipe and Split-Screen controls to compare clips and make whatever adjustments are necessary to make each one match the main look for the scene.



Scene balancing adjustments made using the Clip mode

Once all the clips are balanced with one another, you can choose Post-Clip Group from the Node Editor mode drop-down. This lets you add one more grade on top of the Pre-Clip and Clip grades that you've already applied, that you can use for any number of things. In this example, a single stylistic adjustment is applied that now affects all of the clips at once. If a client wants to see something different, you can easily make the necessary changes to one clip, and all other clips in that group will immediately be altered as well.



A stylistic adjustment applied to the Group Post-Clip mode

At this point, the scene is graded via a well organized set of corrections. If the client later wants a change affecting the underlying primary grade that everything is built on, you can adjust the Pre-Clip grade. If you spot an inconsistency with your shot matching at any point, you can make a fast tweak to the relevant Clip grade. And if the client wants a stylistic change, you can make any necessary adjustments to the Post-Clip grade to change the overall look of the whole scene. In each case, groups give you total control over which adjustments will ripple across the whole group, and which adjustments will be clip-specific.

## Using Undo in Groups

Each Node Editor mode has a separate undo stack, meaning that separate multiple-level undo is saved for Group Pre-Clip, Clip, and Group Post-Clip.

## Saving Stills and Grades in Groups

When you save a still for a clip that's part of a group, the result is that the still reflects the look of the combined Pre-Clip, Clip, and Post-Clip node trees, but the grade that's saved within depends on the Node Editor mode you had selected when saving the still. For example, if the Node Editor is set to Group Pre-Clip, then you'll only save the Pre-Clip grade; the Clip, Group Post-Clip, and Track grades will be ignored. Copying a saved grade to a clip in a group results in that grade being copied to the node tree of whichever Node Editor mode is currently open.

## Collapse Group Grades

If you want to take a clip out of a group, but you want its grade to continue incorporating all adjustments made in the Pre-Group and Post-Group Node Editor modes, you can use the Collapse Group Grades command to copy all nodes in the Pre-Group and Post-Group grades to the Clip grade. When you do this, Pre-Group nodes are added before any pre-existing nodes in the Clip grade, and all Post-Group nodes are added after, in order to maintain the correct order of operations.



(Top) The Clip node tree, (Bottom) After using the Collapse Group Grades command to flatten the Pre-Group and Post-Group nodes into the Clip node tree

**To flatten all group grades into a single Clip mode node tree:**

Right-click a clip's thumbnail in the Thumbnail timeline, and choose Collapse Group Grades from the contextual menu.

Using Collapse Group Grades on a clip always removes that clip from whatever group it was previously a member of. This can also be an easy way of creating a single flat node tree in preparation for saving a grade to the Gallery for applying to other clips that aren't themselves in groups.

## Groups and Versions

Local or Remote versions you create relate only to the Clip grade. Group Pre-Clip and Group Post-Clip grades cannot be versioned.

## Exporting Grades and LUTs

If you find it necessary to exchange grades with other workstations, there are two ways you can do so directly: by exporting grades, or LUTs. This section discusses the export of Grades.

For more information about exporting LUTs, see *Chapter 150, "Using LUTs."*

**To export a grade:**

- 1 Save the grade you want to export as a still in the Gallery.
- 2 Right-click the saved still in the Gallery, and choose Export.
- 3 Select the image format you want to save the still as. Choices include: DPX, Cineon, TIFF, JPEG, PNG, PPM, BMP, and XPM.
- 4 Choose a location for the resulting still image and saved grade files, type a name into the Save As field, and click Save.

Two files are saved. The image file contains the still image of the frame that was stored. A DRX (DaVinci Resolve eXchange) file contains all the grading information.

**To Import a grade:**

- 1 Right-click anywhere in the gray area of the Gallery, and choose Import.
- 2 Click the Options button to select the specific file type you want to import, or select "All Files" for multiple formats.
- 3 Choose the image file that was exported from your DaVinci Resolve workstation, or someone else's. The accompanying DRX file must be in the same location. If you lose the original image file, you can still import the DRX file by itself. It will preserve the node structure of the still, but you will not be able to wipe against it.

The still you imported appears in the Gallery, containing the grading information you wanted to import.